

Newton's formula (in symmetric polynomials)

$$\forall k \geq 0, p_k := \sum_{i=1}^n X_i^k \quad (p_0 = n).$$

- $1 \leq k \leq n$, $\underbrace{p_k}_{e_0 p_k} - e_1 p_{k-1} + e_2 p_{k-2} + \dots + (-1)^k k e_k = 0$.
- $k \geq n$, $p_k - e_1 p_{k-1} + \dots + (-1)^n e_n p_{k-n} = 0$.

Using this, one can

- express $\forall p_k$ in terms of $e_1, \dots, e_n$, with coefficients $/\mathbb{Z}$.
- express $e_1, \dots, e_n$ in terms of $p_1, p_2, \dots$ with coefficients $/\mathbb{Q}$.

Sketch of Proof Consider formal power series in Y :

$$\begin{aligned} P(Y) &:= \sum_{k \geq 1} p_k Y^{k-1} = \sum_{k \geq 1} \sum_{i=1}^n X_i^k Y^{k-1} \\ &= \sum_{i=1}^n \frac{X_i}{1 - X_i Y}, \end{aligned}$$

$$E(Y) := \sum_{k=0}^n e_k Y^k = \prod_{i=1}^n (1 + X_i Y).$$

$$\text{Now } P(-Y) = \sum_{i=1}^n \frac{X_i}{1 + X_i Y} = \frac{E'(Y)}{E(Y)},$$

$$\left(\sum_{a \geq 0} e_a Y^a \right) \left(\sum_{b \geq 1} p_b (-Y)^{b-1} \right) = E(Y) P(-Y) = E'(Y) = \sum_{k=1}^n k e_k Y^{k-1}.$$

$$\text{Coefficient of } Y^{k-1} \text{ is } \sum_{a+b=k} e_a p_b (-1)^{b-1} = k e_k.$$

§ Irreducible polynomials

Recall R : integral domain, $x, y \in R$. If $\exists u \in R^\times, x = uy$, we write $x \sim y$.

Let $x \in R \setminus R^\times, x \neq 0$. If $x = ab \implies a \sim 1 \vee b \sim 1$, we say x is **irreducible** in R .

Now let F : field, $R = F[X], R^\times = F^\times$. $f \in F[X]$ is irreducible means:

$$f = g_1 g_2 \implies g_1 \sim 1 \vee g_2 \sim 1.$$

Things are more complicated on $\mathbb{Q}[X]$ and $\mathbb{Z}[X]$.

Definition Let $f = a_0 + \dots + a_n x_n \in \mathbb{Z}[X] \setminus \{0\}$. $c(f) := \gcd(a_0, \dots, a_n) \in \mathbb{Z}_{\geq 1}$. If $c(f) = 1$, we say f is primitive.

Lemma (Gauss) Let $g, h \in \mathbb{Z}[X]$ be both primitive, then so is gh .

Proof $\forall p$: prime, write

$$\begin{aligned} g &= a_0 + \dots + \underbrace{a_r X^r}_{p \nmid a_r} + \underbrace{\dots}_{p \mid a_i}, \\ h &= b_0 + \dots + \underbrace{b_s X^s}_{p \nmid b_s} + \underbrace{\dots}_{p \mid b_j}. \end{aligned}$$

The coefficient of X^{r+s} in gh is $\sum_{i+j=r+s} a_i b_j \equiv a_r b_s \not\equiv 0 \pmod{p} \implies p \nmid c(gh)$.

Vary p to reach conclusion. □

Alternatively, for any prime p we can argue

$$\begin{aligned}\bar{g} &:= g \bmod p \in \mathbb{F}_p[X] \\ &= \sum_i \bar{a}_i X^i = \bar{a}_0 + \dots + \bar{a}_r X^r \quad (\bar{a}_r \neq 0). \\ \bar{h} &= \bar{b}_0 + \dots + \bar{b}_s X^s.\end{aligned}$$

$$\overline{gh} = \bar{g} \cdot \bar{h} \neq 0, \text{ i.e. } p \nmid c(gh).$$

Lemma Let $g, h \in \mathbb{Z}[X] \setminus \{0\}$, $c(gh) = c(g)c(h)$.

Proof $g = c(g)g_0, h = c(h)h_0, gh = c(g)c(h) \underbrace{g_0 h_0}_{\text{primitive}} \implies c(gh) = c(g)c(h)$. □

Theorem Let $f \in \mathbb{Z}[X] \setminus \{0\}$ be primitive.

- If $f = gh$ in $\mathbb{Q}[X]$, then $\exists \alpha \in \mathbb{Q}^\times$ s.t. $g_1 := \alpha g, h_1 := \alpha^{-1}h$ are both in $\mathbb{Z}[X]$ and primitive.
- When $\deg f > 0$, the following are equivalent:
 - f is irreducible in $\mathbb{Q}[X]$,
 - There are no $g, h \in \mathbb{Z}[X]$ s.t. $f = gh$ and $\deg g, \deg h > 0$.

Proof

- $\deg f = 0 \implies$ take $\alpha := g^{-1}$.

Assume $\deg f > 0, f = gh$ in $\mathbb{Q}[X]$. Take $u, v \in \mathbb{Z}_{\geq 1}$ s.t. $ug, vh \in \mathbb{Z}[X]$.

$$\begin{aligned}\implies uv &= c(uvf) = c(ug \cdot vh) \stackrel{\text{Lem}}{=} c(ug)c(vh) \implies f = \frac{ug}{c(ug)} \cdot \frac{vh}{c(vh)}, \\ \text{i.e. can take } \alpha &:= \frac{u}{c(ug)}.\end{aligned}$$

- $(\neg b \implies \neg a) \quad f = gh$ in $\mathbb{Z}[X], \deg f, \deg h > 0 \implies f$ reducible in $\mathbb{Z}[X]$.

$(\neg a \implies \neg b) \quad$ Suppose $f = gh, g, h \in \mathbb{Q}[X], \deg > 0$. By (1) $f = g_1 h_1 \in \mathbb{Z}[X]$. □

Recall Let R : integral domain. We say R is **UFD** if $\forall r \in R \setminus \{0\}$ has a decomposition

$$r \sim p_1 \dots p_m, \quad p_i \text{ irreducible in } R \quad (m \in \mathbb{Z}_{\geq 0})$$

and $p_1, \dots, p_m$ are unique up to permutations.

Theorem' $\mathbb{Z}[X]$ is UFD, whose irreducibles are

- (Type I) Irreducibles in $\mathbb{Z}$,
- (Type II) Primitive $f \in \mathbb{Z}[X]$ satisfying the property (2) in Thm.

Proof Firstly we classify the irreducibles. Type I are the irreducibles of $\mathbb{Z}[X]$ with $\deg = 0$.

Now assume $f \in \mathbb{Z}[X], \deg f > 0, f = c(f)f_0$.

If f is irreducible in $\mathbb{Z}[X]$, then $c(f) = 1$ (otherwise can take some integer factor) $\implies f$ primitive.

If $f = gh$ in $\mathbb{Z}[X], g, h \notin \mathbb{Z}^\times$.

$$f \text{ primitive} \implies \deg g, \deg h > 0.$$

So f is irreducible in $\mathbb{Z}[X] \iff$ there are no $g, h \in \mathbb{Z}[X]$ s.t. $f = gh, \deg g, \deg h > 0$.

Thm gives the conclusion that Type II are exactly the irreducibles in $\mathbb{Z}[X]$ of $\deg > 0$.

Now we will prove that $\mathbb{Z}[X]$ is UFD. Let $f \in \mathbb{Z}[X] \setminus \{0\}$. Firstly, $f = c(f)f_0$ where $c(f)$ is the product of irreducibles of type I ($\because \mathbb{Z}$ is a UFD). Then decompose f_0 into irreducibles in $\mathbb{Q}[X]$. Thm: may assume (up to $\mathbb{Q}^\times$) that each factor is primitive in $\mathbb{Z}[X]$, i.e. irreducibles in $\mathbb{Z}[X]$ of type II.

$\Rightarrow f \sim$ product of irreducibles in $\mathbb{Z}[X]$.

Uniqueness: $\underbrace{a_1 \dots a_m}_I \underbrace{p_1 \dots p_n}_II \sim \underbrace{b_1 \dots b_r}_I \underbrace{q_1 \dots q_s}_II$.

By taking $c(\cdot)$ we can show $\prod a_i \sim \prod b_j$ in $\mathbb{Z}$, hence $\prod p_i \sim \prod q_j$. We already know $F : \text{field} \Rightarrow F[X] : \text{UFD}$, hence after converting p_i, q_j to primitive polynomials on $\mathbb{Z}[X]$, two decompositions are equal up to $\sim$ and permutation.

Remark We used tools on $\mathbb{Q}[X]$ to prove the $\mathbb{Z}$ version. The other way is also possible by multiplying a common integer.